



**Generation Interconnection
System Impact Study Report
for**

Queue Project AF2-107

CLIFFORD 138 KV

32.3 MW Capacity / 50 MW Energy

February 2021

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is AEP.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Amherst County, Virginia. The installed facilities will have a total capability of 50 MW with 32.3 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is February 01, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-107
Project Name	CLIFFORD 138 KV
State	Virginia
County	Amherst
Transmission Owner	AEP
MFO	50
MWE	50
MWC	32.3
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF2-107 will interconnect with the AEP transmission system via a direct connection to the Clifford 138 kV station.

To accommodate the interconnection to the Clifford 138 kV station, the substation will have to be expanded requiring the rebuild of the 138 kV bus, installation of three (3) new 138 kV circuit breakers, physically constructed and operated as a ring bus (see Attachment 1). Installation of associated protection and control equipment, line risers, SCADA, jumpers, switches, and 138 kV revenue metering will also be required. AEP reserves the right to specify the final acceptable configuration considering design practices, future expansion, and compliance requirements.

AEP will extend one span of 138 kV transmission line for the generation-leads going to the AF2-107 site. Unless this span extends directly from within the AEP station at the POI to the IC collector station structure, AEP will build and own the first transmission line structure outside of the Clifford 138 kV station fence to which the AEP and AF2-107 transmission line conductors will attach.

5 Cost Summary

The AF2-107 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$9,061,000
Allocation towards System Network Upgrade Costs*	\$5,353,000
Total Costs	\$14,414,000

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

The estimates provided in this report are preliminary in nature, as they were determined without the benefit of detailed engineering studies. Final estimates will require an on-site review and coordination to determine final construction requirements. In addition, Stability analysis will be completed during the Facilities Study stage. It is possible that a need for additional upgrades could be identified by these studies.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost

allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The total physical interconnection costs is given in the table below:

6.1 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
138 kV Revenue Metering	\$376,000
Generator lead first span exiting the POI station, including the first structure outside the fence	\$400,000
Total Attachment Facility Costs	\$776,000

6.2 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Three (3) new 138 kV circuit breakers will be installed at the Clifford 138 kV station. Installation of associated protection and control equipment, 138 kV line risers, and SCADA will also be required.	\$8,150,000
Total Direct Connection Facility Costs	\$8,150,000

6.3 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Upgrade Line Protections & Controls at the Reusens 138 kV Substation	\$45,000
Upgrade Line Protections & Controls at the Scottsville 138 kV Substation	\$45,000
Upgrade Line Protections & Controls at the Clifford 138 kV Substation	\$45,000
Total Non-Direct Connection Facility Costs	\$135,000

7 Schedule

It is anticipated that the time between receipt of executed Agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would generally be between 24 to 36 months after Agreement execution.

8 Interconnection Customer Requirements

It is understood that the Interconnection Customer (IC) is responsible for all costs associated with this interconnection. The costs above are reimbursable to the Transmission Owner. The cost of the IC's generating plant and the costs for the line connecting the generating plant to the Point of Interconnection are not included in this report; these are assumed to be the IC's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for the Transmission Owner to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Irradiance (Watts/meter²) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Ambient air temperature (Fahrenheit) - (Accepted, not required)
- Wind speed (meters/second) - (Accepted, not required)
- Wind direction (decimal degrees from true north) - (Accepted, not required)

9.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

10 Summer Peak Analysis

The Queue Project AF2-107 was evaluated as a 50.0 MW (Capacity 32.3 MW) injection at the Clifford 138 kV substation in the AEP area. Project AF2-107 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-107 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

10.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
97937263	313707	6FORK UNION	230.0	DVP	961060	AF2-397 TAP	230.0	DVP	1	DVP_P1-2: LN 2027-A	single	661.760009766	99.82	100.71	AC	6.06
97937264	313707	6FORK UNION	230.0	DVP	961060	AF2-397 TAP	230.0	DVP	1	3138686CARTER V 230 933500 AC2-165 TAP 230 1	single	661.760009766	99.41	100.3	AC	6.06

10.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

10.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
97730835	313707	6FORK UNION	230.0	DVP	961060	AF2-397 TAP	230.0	DVP	1	DVP_P4-2: 202712	breaker	809.0	104.29	105.41	AC	9.38
97730796	314765	6MTEAGLE	230.0	DVP	314749	6CHARLVL	230.0	DVP	1	DVP_P4-2: 202712	breaker	809.0	100.02	101.14	AC	9.38
97730775	961060	AF2-397 TAP	230.0	DVP	314765	6MTEAGLE	230.0	DVP	1	DVP_P4-2: 202712	breaker	809.0	103.52	104.64	AC	9.38

10.4 Steady-State Voltage Requirements

None

10.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed

with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
95684933	242603	05CLIFFR	138.0	AEP	242563	05BOXWD	138.0	AEP	1	AEP_P1-2_#10336-B	operation	167.0	87.0	109.91	AC	41.82
97937256	313707	6FORK UNION	230.0	DVP	961060	AF2-397 TAP	230.0	DVP	1	DVP_P1-2: LN 2027-B	operation	661.760009766	120.62	121.99	AC	9.38
97937231	314765	6MTEAGLE	230.0	DVP	314749	6CHARLV L	230.0	DVP	1	DVP_P1-2: LN 2027-B	operation	661.760009766	107.53	108.9	AC	9.38
97194636	938820	AE1-108 TAP	138.0	DVP	314746	4BREMO	138.0	DVP	1	AEP_SUBT_P1 - 3_#231_05CLIFFR 138_1	operation	156.979995728	108.61	139.81	AC	50.0
97194637	938820	AE1-108 TAP	138.0	DVP	314746	4BREMO	138.0	DVP	1	AEP_P1-2_#5400	operation	156.979995728	108.61	139.81	AC	50.0
97194752	941010	AE2-092 TAP	115.0	DVP	314774	3SHERW OD	115.0	DVP	1	DVP_P1-2: LN 2028-B	operation	169.199996948	109.28	110.71	AC	2.87
97937185	961060	AF2-397 TAP	230.0	DVP	314765	6MTEAGLE	230.0	DVP	1	DVP_P1-2: LN 2027-B	operation	661.760009766	111.8	113.17	AC	9.38

10.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-107	Upgrade Number																								
97730796	2	6MTEAGLE 230.0 kV - 6CHARLVL 230.0 kV Ckt 1	Rebuild 6.47 miles of 230 kV Line 2028 from Mt Eagle to Charlottesville with 2-636 ACSR. New expected SE rating to be 1047 MVA. Time estimate: 30-36 months. The cost allocation is as follows: <table><thead><tr><th>Queue</th><th>MW contribution</th><th>% of Cost</th><th>Cost (\$16.175 M)</th></tr></thead><tbody><tr><td>AF1-293</td><td>37.7</td><td>58.27</td><td>9.425</td></tr><tr><td>AF1-294</td><td>6.1</td><td>9.43</td><td>1.525</td></tr><tr><td>AF1-323</td><td>11.5</td><td>17.77</td><td>2.875</td></tr><tr><td>AF2-107</td><td>9.4</td><td>14.53</td><td>2.350</td></tr></tbody></table>	Queue	MW contribution	% of Cost	Cost (\$16.175 M)	AF1-293	37.7	58.27	9.425	AF1-294	6.1	9.43	1.525	AF1-323	11.5	17.77	2.875	AF2-107	9.4	14.53	2.350	\$16.175 M	\$2.35 M	N6757				
Queue	MW contribution	% of Cost	Cost (\$16.175 M)																											
AF1-293	37.7	58.27	9.425																											
AF1-294	6.1	9.43	1.525																											
AF1-323	11.5	17.77	2.875																											
AF2-107	9.4	14.53	2.350																											
97730775	3	AF2-397 TAP 230.0 kV - 6MTEAGLE 230.0 kV Ckt 1	Rebuild 17.42 miles of 230 kV Line 2028 from Fork Union to Mount Eagle with 2-636 ACSR. New expected SE/SLD ratings to be 1047/1204 MVA. Cost estimate: \$26.13 M. Time estimate: 36-40 months. The cost allocation is: <table><thead><tr><th>Queue</th><th>MW contribution</th><th>% of Cost</th><th>Cost (\$26.130 M)</th></tr></thead><tbody><tr><td>AF1-059</td><td>7.8</td><td>9.54</td><td>2.492</td></tr><tr><td>AF1-293</td><td>47.0</td><td>57.46</td><td>15.014</td></tr><tr><td>AF1-294</td><td>6.1</td><td>7.46</td><td>1.949</td></tr><tr><td>AF1-323</td><td>11.5</td><td>14.06</td><td>3.674</td></tr><tr><td>AF2-107</td><td>9.4</td><td>11.49</td><td>3.003</td></tr></tbody></table>	Queue	MW contribution	% of Cost	Cost (\$26.130 M)	AF1-059	7.8	9.54	2.492	AF1-293	47.0	57.46	15.014	AF1-294	6.1	7.46	1.949	AF1-323	11.5	14.06	3.674	AF2-107	9.4	11.49	3.003	\$26.13 M	\$3.003 M	N6758
Queue	MW contribution	% of Cost	Cost (\$26.130 M)																											
AF1-059	7.8	9.54	2.492																											
AF1-293	47.0	57.46	15.014																											
AF1-294	6.1	7.46	1.949																											
AF1-323	11.5	14.06	3.674																											
AF2-107	9.4	11.49	3.003																											
97937264,97730835,97937263	1	6FORK UNION 230.0 kV - AF2-397 TAP 230.0 kV Ckt 1	Same as AF2-397 TAP 230.0 kV - 6MTEAGLE 230.0 kV Ckt 1 above.																											

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

10.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

10.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
97730835	313707	6FORK UNION	DVP	961060	AF2-397 TAP	DVP	1	DVP_P4-2: 202712	breaker	809.0	104.29	105.41	AC	9.38

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.1328	50/50	0.1328
314429	3JTRSVLE	0.5312	50/50	0.5312
314677	6BUCKING	1.0418	50/50	1.0418
315191	1BEARGRDN G1	14.6949	50/50	14.6949
315192	1BEARGRDN G2	14.6949	50/50	14.6949
315193	1BEARGRDN S1	30.2431	50/50	30.2431
920291	AA2-127	3.0527	50/50	3.0527
924021	AB2-043 C O1	0.4201	50/50	0.4201
924022	AB2-043 E O1	2.2355	50/50	2.2355
924032	AB2-045 E	2.8932	50/50	2.8932
924161	AB2-060 C OP	1.2430	50/50	1.2430
924162	AB2-060 E OP	3.4811	50/50	3.4811
924301	AB2-077 C O1 (Suspended)	1.5500	50/50	1.5500
924302	AB2-077 E O1 (Suspended)	1.0334	50/50	1.0334
924311	AB2-078 C O1 (Suspended)	1.5500	50/50	1.5500
924312	AB2-078 E O1 (Suspended)	1.0334	50/50	1.0334
924321	AB2-079 C O1 (Suspended)	1.5500	50/50	1.5500
924322	AB2-079 E O1 (Suspended)	1.0334	50/50	1.0334
925611	AC1-036 C	1.1045	50/50	1.1045
925612	AC1-036 E	1.8022	50/50	1.8022
925781	AC1-054 C O1	3.1043	Adder	3.65
925782	AC1-054 E O1	1.4301	Adder	1.68
925831	AC1-062	0.0788	50/50	0.0788
926271	AC1-105 C O1 (Suspended)	2.7021	Adder	3.18
926272	AC1-105 E O1 (Suspended)	1.3451	Adder	1.58
926451	AC1-116 C	-0.9024	Adder	-1.06
932511	AC2-071 C	6.1079	50/50	6.1079
932512	AC2-071 E	3.0769	50/50	3.0769
933501	AC2-165 C	32.1577	50/50	32.1577
933502	AC2-165 E	24.2029	50/50	24.2029
934611	AD1-087 C O1	3.5747	Adder	4.21
934612	AD1-087 E O1	1.6800	Adder	1.98
934621	AD1-088 C	19.9927	50/50	19.9927
934622	AD1-088 E	9.4083	50/50	9.4083

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
935171	AD1-152 C O1	3.5525	Adder	4.18
935172	AD1-152 E O1	2.3683	Adder	2.79
935221	AD1-157 C	2.3855	50/50	2.3855
935222	AD1-157 E	1.5904	50/50	1.5904
936261	AD2-033 C	14.2467	50/50	14.2467
936262	AD2-033 E	9.4978	50/50	9.4978
936361	AD2-046 C O1	4.6426	Adder	5.46
936362	AD2-046 E O1	2.1349	Adder	2.51
936481	AD2-063 C O1	14.0373	50/50	14.0373
936482	AD2-063 E O1	9.2802	50/50	9.2802
937481	AD2-202 C O1	0.9473	Adder	1.11
937482	AD2-202 E O1	0.5329	Adder	0.63
938371	AE1-056 C	10.2843	50/50	10.2843
938372	AE1-056 E	5.6193	50/50	5.6193
938821	AE1-108 C	35.7992	50/50	35.7992
938822	AE1-108 E	23.8662	50/50	23.8662
939181	AE1-148 C O1	5.4724	50/50	5.4724
939182	AE1-148 E O1	3.6482	50/50	3.6482
940241	AE2-006	0.5620	50/50	0.5620
940661	AE2-053 O1	2.0268	50/50	2.0268
941011	AE2-092 C	29.1920	50/50	29.1920
941012	AE2-092 E	19.4613	50/50	19.4613
941791	AE2-182 C	3.2481	50/50	3.2481
941792	AE2-182 E	1.5121	50/50	1.5121
942451	AE2-258	1.8651	50/50	1.8651
942461	AE2-259 C O1	18.9534	50/50	18.9534
942462	AE2-259 E O1	12.6356	50/50	12.6356
943901	AF1-058 C	0.8487	Adder	1.0
943902	AF1-058 E	0.5658	Adder	0.67
943911	AF1-059	8.1583	Adder	9.6
946291	AF1-293 C O1	47.0499	50/50	47.0499
946292	AF1-293 E O1	8.4638	50/50	8.4638
946301	AF1-294 C	6.0752	50/50	6.0752
946302	AF1-294 E	4.0501	50/50	4.0501
946591	AF1-323 C	11.4718	50/50	11.4718
946592	AF1-323 E	7.6479	50/50	7.6479
958131	AF2-107 C	6.0595	50/50	6.0595
958132	AF2-107 E	3.3205	50/50	3.3205
958211	AF2-115 C	3.7044	50/50	3.7044
958212	AF2-115 E	2.4696	50/50	2.4696
958801	AF2-171 C	22.0797	50/50	22.0797
958802	AF2-171 E	14.7198	50/50	14.7198
959311	AF2-222 C	20.1290	50/50	20.1290
959312	AF2-222 E	13.4864	50/50	13.4864
959751	AF2-266	4.0983	Adder	4.82
960061	AF2-297 C	3.3950	Adder	3.99
960062	AF2-297 E	2.2633	Adder	2.66
960111	AF2-302 C	4.1716	50/50	4.1716
960112	AF2-302 E	2.7810	50/50	2.7810
961063	AF2-397 BAT	19.0219	50/50	19.0219
WEC	WEC	0.1251	Confirmed LTF	0.1251
LGEE	LGEE	0.2482	Confirmed LTF	0.2482

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CPLE	CPLE	2.1710	Confirmed LTF	2.1710
CBM-W2	CBM-W2	6.5766	Confirmed LTF	6.5766
NY	NY	0.4280	Confirmed LTF	0.4280
TVA	TVA	1.4980	Confirmed LTF	1.4980
O-066	O-066	5.5709	Confirmed LTF	5.5709
CBM-S2	CBM-S2	14.4558	Confirmed LTF	14.4558
CBM-S1	CBM-S1	8.1110	Confirmed LTF	8.1110
G-007	G-007	0.8653	Confirmed LTF	0.8653
MADISON	MADISON	0.5463	Confirmed LTF	0.5463
MEC	MEC	0.8692	Confirmed LTF	0.8692
AA2-074	AA2-074	1.4783	LTF	1.4783
CBM-W1	CBM-W1	4.3910	Confirmed LTF	4.3910

10.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
97730796	314765	6MTEAGLE	DVP	314749	6CHARLVL	DVP	1	DVP_P4-2: 202712	breaker	809.0	100.02	101.14	AC	9.38

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.1328	50/50	0.1328
314429	3JTRSVLE	0.5312	50/50	0.5312
314677	6BUCKING	1.0418	50/50	1.0418
315191	1BEARGRDN G1	14.6949	50/50	14.6949
315192	1BEARGRDN G2	14.6949	50/50	14.6949
315193	1BEARGRDN S1	30.2431	50/50	30.2431
920291	AA2-127	3.0527	50/50	3.0527
924021	AB2-043 C O1	0.4201	50/50	0.4201
924022	AB2-043 E O1	2.2355	50/50	2.2355
924032	AB2-045 E	2.8932	50/50	2.8932
924161	AB2-060 C OP	1.2430	50/50	1.2430
924162	AB2-060 E OP	3.4811	50/50	3.4811
924301	AB2-077 C O1 (Suspended)	1.5500	50/50	1.5500
924302	AB2-077 E O1 (Suspended)	1.0334	50/50	1.0334
924311	AB2-078 C O1 (Suspended)	1.5500	50/50	1.5500
924312	AB2-078 E O1 (Suspended)	1.0334	50/50	1.0334
924321	AB2-079 C O1 (Suspended)	1.5500	50/50	1.5500
924322	AB2-079 E O1 (Suspended)	1.0334	50/50	1.0334
925611	AC1-036 C	1.1045	50/50	1.1045
925612	AC1-036 E	1.8022	50/50	1.8022
925831	AC1-062	0.0788	50/50	0.0788
926271	AC1-105 C O1 (Suspended)	2.7021	Adder	3.18
926272	AC1-105 E O1 (Suspended)	1.3451	Adder	1.58
926451	AC1-116 C	3.6384	50/50	3.6384
926452	AC1-116 E	6.0382	50/50	6.0382
932511	AC2-071 C	6.1079	50/50	6.1079
932512	AC2-071 E	3.0769	50/50	3.0769
933501	AC2-165 C	32.1577	50/50	32.1577
933502	AC2-165 E	24.2029	50/50	24.2029
934621	AD1-088 C	19.9927	50/50	19.9927
934622	AD1-088 E	9.4083	50/50	9.4083
935221	AD1-157 C	2.3855	50/50	2.3855
935222	AD1-157 E	1.5904	50/50	1.5904
936261	AD2-033 C	14.2467	50/50	14.2467

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
936262	AD2-033 E	9.4978	50/50	9.4978
936361	AD2-046 C O1	4.6426	Adder	5.46
936362	AD2-046 E O1	2.1349	Adder	2.51
936481	AD2-063 C O1	14.0373	50/50	14.0373
936482	AD2-063 E O1	9.2802	50/50	9.2802
938371	AE1-056 C	10.2843	50/50	10.2843
938372	AE1-056 E	5.6193	50/50	5.6193
938821	AE1-108 C	35.7992	50/50	35.7992
938822	AE1-108 E	23.8662	50/50	23.8662
939181	AE1-148 C O1	5.4724	50/50	5.4724
939182	AE1-148 E O1	3.6482	50/50	3.6482
940241	AE2-006	0.5620	50/50	0.5620
940661	AE2-053 O1	2.0268	50/50	2.0268
941011	AE2-092 C	29.1920	50/50	29.1920
941012	AE2-092 E	19.4613	50/50	19.4613
941791	AE2-182 C	3.2481	50/50	3.2481
941792	AE2-182 E	1.5121	50/50	1.5121
942451	AE2-258	1.8651	50/50	1.8651
942461	AE2-259 C O1	18.9534	50/50	18.9534
942462	AE2-259 E O1	12.6356	50/50	12.6356
943911	AF1-059	8.1583	Adder	9.6
946291	AF1-293 C O1	47.0499	50/50	47.0499
946292	AF1-293 E O1	8.4638	50/50	8.4638
946301	AF1-294 C	6.0752	50/50	6.0752
946302	AF1-294 E	4.0501	50/50	4.0501
946591	AF1-323 C	11.4718	50/50	11.4718
946592	AF1-323 E	7.6479	50/50	7.6479
958131	AF2-107 C	6.0595	50/50	6.0595
958132	AF2-107 E	3.3205	50/50	3.3205
958211	AF2-115 C	3.7044	50/50	3.7044
958212	AF2-115 E	2.4696	50/50	2.4696
958801	AF2-171 C	22.0797	50/50	22.0797
958802	AF2-171 E	14.7198	50/50	14.7198
959311	AF2-222 C	20.1290	50/50	20.1290
959312	AF2-222 E	13.4864	50/50	13.4864
959751	AF2-266	4.0983	Adder	4.82
960111	AF2-302 C	4.1716	50/50	4.1716
960112	AF2-302 E	2.7810	50/50	2.7810
961061	AF2-397 C	49.8194	50/50	49.8194
961062	AF2-397 E	33.2129	50/50	33.2129
WEC	WEC	0.1251	Confirmed LTF	0.1251
LGEE	LGEE	0.2482	Confirmed LTF	0.2482
CPL	CPL	2.1710	Confirmed LTF	2.1710
CBM-W2	CBM-W2	6.5766	Confirmed LTF	6.5766
NY	NY	0.4280	Confirmed LTF	0.4280
TVA	TVA	1.4980	Confirmed LTF	1.4980
O-066	O-066	5.5709	Confirmed LTF	5.5709
CBM-S2	CBM-S2	14.4558	Confirmed LTF	14.4558
CBM-S1	CBM-S1	8.1110	Confirmed LTF	8.1110
G-007	G-007	0.8653	Confirmed LTF	0.8653
MADISON	MADISON	0.5463	Confirmed LTF	0.5463
MEC	MEC	0.8692	Confirmed LTF	0.8692

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
AA2-074	AA2-074	1.4783	LTF	1.4783
CBM-W1	CBM-W1	4.3910	Confirmed LTF	4.3910

10.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
97730775	961060	AF2-397 TAP	DVP	314765	6MTEAGLE	DVP	1	DVP_P4-2: 202712	breaker	809.0	103.52	104.64	AC	9.38

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.1328	50/50	0.1328
314429	3JTRSVLE	0.5312	50/50	0.5312
314677	6BUCKING	1.0418	50/50	1.0418
315191	1BEARGRDN G1	14.6949	50/50	14.6949
315192	1BEARGRDN G2	14.6949	50/50	14.6949
315193	1BEARGRDN S1	30.2431	50/50	30.2431
920291	AA2-127	3.0527	50/50	3.0527
924021	AB2-043 C O1	0.4201	50/50	0.4201
924022	AB2-043 E O1	2.2355	50/50	2.2355
924032	AB2-045 E	2.8932	50/50	2.8932
924161	AB2-060 C OP	1.2430	50/50	1.2430
924162	AB2-060 E OP	3.4811	50/50	3.4811
924301	AB2-077 C O1 (Suspended)	1.5500	50/50	1.5500
924302	AB2-077 E O1 (Suspended)	1.0334	50/50	1.0334
924311	AB2-078 C O1 (Suspended)	1.5500	50/50	1.5500
924312	AB2-078 E O1 (Suspended)	1.0334	50/50	1.0334
924321	AB2-079 C O1 (Suspended)	1.5500	50/50	1.5500
924322	AB2-079 E O1 (Suspended)	1.0334	50/50	1.0334
925611	AC1-036 C	1.1045	50/50	1.1045
925612	AC1-036 E	1.8022	50/50	1.8022
925831	AC1-062	0.0788	50/50	0.0788
926271	AC1-105 C O1 (Suspended)	2.7021	Adder	3.18
926272	AC1-105 E O1 (Suspended)	1.3451	Adder	1.58
926451	AC1-116 C	-0.9024	Adder	-1.06
932511	AC2-071 C	6.1079	50/50	6.1079
932512	AC2-071 E	3.0769	50/50	3.0769
933501	AC2-165 C	32.1577	50/50	32.1577
933502	AC2-165 E	24.2029	50/50	24.2029
934611	AD1-087 C O1	3.5747	Adder	4.21
934612	AD1-087 E O1	1.6800	Adder	1.98
934621	AD1-088 C	19.9927	50/50	19.9927
934622	AD1-088 E	9.4083	50/50	9.4083
935171	AD1-152 C O1	3.5525	Adder	4.18
935172	AD1-152 E O1	2.3683	Adder	2.79

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
935221	AD1-157 C	2.3855	50/50	2.3855
935222	AD1-157 E	1.5904	50/50	1.5904
936261	AD2-033 C	14.2467	50/50	14.2467
936262	AD2-033 E	9.4978	50/50	9.4978
936361	AD2-046 C O1	4.6426	Adder	5.46
936362	AD2-046 E O1	2.1349	Adder	2.51
936481	AD2-063 C O1	14.0373	50/50	14.0373
936482	AD2-063 E O1	9.2802	50/50	9.2802
937481	AD2-202 C O1	0.9473	Adder	1.11
937482	AD2-202 E O1	0.5329	Adder	0.63
938371	AE1-056 C	10.2843	50/50	10.2843
938372	AE1-056 E	5.6193	50/50	5.6193
938821	AE1-108 C	35.7992	50/50	35.7992
938822	AE1-108 E	23.8662	50/50	23.8662
939181	AE1-148 C O1	5.4724	50/50	5.4724
939182	AE1-148 E O1	3.6482	50/50	3.6482
940241	AE2-006	0.5620	50/50	0.5620
940661	AE2-053 O1	2.0268	50/50	2.0268
941011	AE2-092 C	29.1920	50/50	29.1920
941012	AE2-092 E	19.4613	50/50	19.4613
941791	AE2-182 C	3.2481	50/50	3.2481
941792	AE2-182 E	1.5121	50/50	1.5121
942451	AE2-258	1.8651	50/50	1.8651
942461	AE2-259 C O1	18.9534	50/50	18.9534
942462	AE2-259 E O1	12.6356	50/50	12.6356
943911	AF1-059	8.1583	Adder	9.6
946291	AF1-293 C O1	47.0499	50/50	47.0499
946292	AF1-293 E O1	8.4638	50/50	8.4638
946301	AF1-294 C	6.0752	50/50	6.0752
946302	AF1-294 E	4.0501	50/50	4.0501
946591	AF1-323 C	11.4718	50/50	11.4718
946592	AF1-323 E	7.6479	50/50	7.6479
958131	AF2-107 C	6.0595	50/50	6.0595
958132	AF2-107 E	3.3205	50/50	3.3205
958211	AF2-115 C	3.7044	50/50	3.7044
958212	AF2-115 E	2.4696	50/50	2.4696
958801	AF2-171 C	22.0797	50/50	22.0797
958802	AF2-171 E	14.7198	50/50	14.7198
959311	AF2-222 C	20.1290	50/50	20.1290
959312	AF2-222 E	13.4864	50/50	13.4864
959751	AF2-266	4.0983	Adder	4.82
960111	AF2-302 C	4.1716	50/50	4.1716
960112	AF2-302 E	2.7810	50/50	2.7810
961061	AF2-397 C	49.8194	50/50	49.8194
961062	AF2-397 E	33.2129	50/50	33.2129
WEC	WEC	0.1251	Confirmed LTF	0.1251
LGEE	LGEE	0.2482	Confirmed LTF	0.2482
CPL	CPL	2.1710	Confirmed LTF	2.1710
CBM-W2	CBM-W2	6.5766	Confirmed LTF	6.5766
NY	NY	0.4280	Confirmed LTF	0.4280
TVA	TVA	1.4980	Confirmed LTF	1.4980
O-066	O-066	5.5709	Confirmed LTF	5.5709

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CBM-S2	CBM-S2	14.4558	Confirmed LTF	14.4558
CBM-S1	CBM-S1	8.1110	Confirmed LTF	8.1110
G-007	G-007	0.8653	Confirmed LTF	0.8653
MADISON	MADISON	0.5463	Confirmed LTF	0.5463
MEC	MEC	0.8692	Confirmed LTF	0.8692
AA2-074	AA2-074	1.4783	LTF	1.4783
CBM-W1	CBM-W1	4.3910	Confirmed LTF	4.3910

10.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA2-074	CPL-PJM	Confirmed
AA2-127	Bear Garden 230kV	In Service
AB2-043	Chase City 115kV	Under Construction
AB2-045	Buckingham 34.5kV	In Service
AB2-060	Chase City-Lunenburg 115kV	In Service
AB2-077	Buggs Island-Chase City 115kV	Suspended
AB2-078	Buggs Island-Chase City 115kV	Suspended
AB2-079	Buggs Island-Chase City 115kV	Suspended
AC1-036	Twittys Creek 34.5kV	Under Construction
AC1-054	Kerr Dam-Eatons Ferry 115 kV	Engineering and Procurement
AC1-062	Jetersville 34.5kV	In Service
AC1-105	Halifax-Mt. Laurel 115kV	Suspended
AC1-116	Mount Eagle 34.5kV	Engineering and Procurement
AC2-071	Buckingham 35kV	Engineering and Procurement
AC2-165	Bremo-Powhatan 230kV	Active
AD1-087	Clover-Sedge Hill 230 kV	Active
AD1-088	Briery-Clover 230 kV	Active
AD1-152	Clover-Sedge Hill 230 kV	Active
AD1-157	South Creek 34.5 kV	Engineering and Procurement
AD2-033	Chase City-Lunenburg 115 kV	Active
AD2-046	Boydton DP-Kerr Dam 115 kV	Active
AD2-063	Central-Chase City 115kV	Active
AD2-202	Clover-Sedge Hill 230kV	Active
AE1-056	Red House-South Creek 115 kV	Active
AE1-108	Bremo-Scottsville 138 kV	Active
AE1-148	Kerr Dam-Ridge Rd 115 kV	Active
AE2-006	Twittys Creek 34.5 kV	Under Construction
AE2-053	Kerr Dam-Ridge Road 115 kV	Active
AE2-092	Kidds Store-Sherwood 115 kV	Active
AE2-182	Briery-Clover 230 kV	Active
AE2-258	Chase City 115 kV	Active
AE2-259	Curdsville-Willis Mtn 115 kV	Active
AF1-058	Welco 34.5 kV	Engineering and Procurement
AF1-059	Brodnax-South Hill 115 kV	Active
AF1-293	Kidds Store-Fort Union 115 kV	Active
AF1-294	Jetersville-Ponton 115 kV	Active
AF1-323	Scottsville-Colleen 138 kV	Active
AF2-107	Clifford 138 kV	Active
AF2-115	Jetersville-Ponton 115 kV	Active

Queue Number	Project Name	Status
AF2-171	Madisonville 115 kV	Active
AF2-222	Madisonville DP-Twitty's Creek 115 kV	Active
AF2-266	Clover 230 kV	Active
AF2-297	Sedge Hill 115 kV	Active
AF2-302	Scottsville-Colleen 138 kV	Active
AF2-397	Fork Union-Mt. Eagle 230 kV	Active

10.9 Contingency Descriptions

Contingency Name	Contingency Definition
AEP_P1-2_#10336-B	CONTINGENCY 'AEP_P1-2_#10336-B' OPEN BRANCH FROM BUS 938820 TO BUS 314746 CKT 1 / 938820 AE1-108 TAP 138 314746 4BREMO 138 1 OPEN BRANCH FROM BUS 242792 TO BUS 243803 CKT 3 / 242792 05SCOTSV 138 243803 05SCOTTVIL 46.0 3 END
AEP_SUBT_P1-3_#231_05CLIFFR 138_1	CONTINGENCY 'AEP_SUBT_P1-3_#231_05CLIFFR 138_1' OPEN BRANCH FROM BUS 242500 TO BUS 242603 CKT 1 / 242500 05CLIFRDEQ 999 242603 05CLIFFR 138 1 OPEN BRANCH FROM BUS 242500 TO BUS 242867 CKT 1 / 242500 05CLIFRDEQ 999 242867 05CLIFFORD 69.0 1 OPEN BRANCH FROM BUS 242500 TO BUS 242888 CKT 1 / 242500 05CLIFRDEQ 999 242888 05CLIFFORD 46.0 1 OPEN BRANCH FROM BUS 242563 TO BUS 242603 CKT 1 / 242563 05BOXWD 138 242603 05CLIFFR 138 1 OPEN BRANCH FROM BUS 242563 TO BUS 242765 CKT 1 / 242563 05BOXWD 138 242765 05REUSEN 138 1 OPEN BRANCH FROM BUS 242563 TO BUS 242772 CKT 1 / 242563 05BOXWD 138 242772 05RIVERV 138 1 REMOVE SWSHUNT FROM BUS 242772 / 242772 05RIVERV 138 OPEN BRANCH FROM BUS 242862 TO BUS 242867 CKT 1 / 242862 05AMHERST 69.0 242867 05CLIFFORD 69.0 1 END
AEP_P1-2_#5400	CONTINGENCY 'AEP_P1-2_#5400' OPEN BRANCH FROM BUS 242500 TO BUS 242603 CKT 1 / 242500 05CLIFRDEQ 999 242603 05CLIFFR 138 1 OPEN BRANCH FROM BUS 242500 TO BUS 242867 CKT 1 / 242500 05CLIFRDEQ 999 242867 05CLIFFORD 69.0 1 OPEN BRANCH FROM BUS 242500 TO BUS 242888 CKT 1 / 242500 05CLIFRDEQ 999 242888 05CLIFFORD 46.0 1 OPEN BRANCH FROM BUS 242563 TO BUS 242603 CKT 1 / 242563 05BOXWD 138 242603 05CLIFFR 138 1 OPEN BRANCH FROM BUS 242563 TO BUS 242765 CKT 1 / 242563 05BOXWD 138 242765 05REUSEN 138 1 OPEN BRANCH FROM BUS 242563 TO BUS 242772 CKT 1 / 242563 05BOXWD 138 242772 05RIVERV 138 1 REMOVE SWSHUNT FROM BUS 242603 / 242603 05CLIFFR 138 REMOVE SWSHUNT FROM BUS 242772 / 242772 05RIVERV 138 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 2027-A	CONTINGENCY 'DVP_P1-2: LN 2027-A' OPEN BRANCH FROM BUS 313868 TO BUS 933500 CKT 1 /* 6CARTERV 230.00 - AC2-165 TAP 230.00 OPEN BRANCH FROM BUS 313868 TO BUS 314747 CKT 1 /* 6CARTERV 230.00 - 6BREMO 230.00 OPEN BUS 313868 /* ISLAND: 6CARTERV 230.00 END
DVP_P1-2: LN 2027-B	CONTINGENCY 'DVP_P1-2: LN 2027-B' OPEN BRANCH FROM BUS 933500 TO BUS 314333 CKT 1 /* AC2-165 TAP 230.00 - 6POWHATN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314322 CKT 1 /* 6JUDES F 230.00 - 6MDLTHAN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314333 CKT 1 /* 6JUDES F 230.00 - 6POWHATN 230.00 OPEN BUS 314310 /* ISLAND: 6JUDES F 230.00 OPEN BUS 314333 /* ISLAND: 6POWHATN 230.00 OPEN BUS 923862 /* ISLAND: AB2-026 E 230.00 END
313868 6CARTERV 230 933500 AC2-165 TAP 230 1	CONTINGENCY '313868 6CARTERV 230 933500 AC2-165 TAP 230 1' OPEN BRANCH FROM BUS 313868 TO BUS 933500 CKT 1 END
DVP_P1-2: LN 2028-B	CONTINGENCY 'DVP_P1-2: LN 2028-B' OPEN BRANCH FROM BUS 961060 TO BUS 314765 CKT 1 /* AF2-397 TAP 230.00 - 6MTEAGLE 230.00 OPEN BRANCH FROM BUS 314749 TO BUS 314765 CKT 1 /* 6CHARLVL 230.00 - 6MTEAGLE 230.00 OPEN BUS 314765 /* ISLAND: 6MTEAGLE 230.00 OPEN BUS 926451 /* ISLAND: AC1-116 C 230.00 OPEN BUS 926452 /* ISLAND: AC1-116 E 230.00 END
Base Case	
DVP_P4-2: 202712	CONTINGENCY 'DVP_P4-2: 202712' /* MIDLOTHIAN 500 230 KV OPEN BRANCH FROM BUS 933500 TO BUS 314333 CKT 1 /* AC2-165 TAP 230.00 - 6POWHATN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314322 CKT 1 /* 6JUDES F 230.00 - 6MDLTHAN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314333 CKT 1 /* 6JUDES F 230.00 - 6POWHATN 230.00 OPEN BUS 314310 /* ISLAND: 6JUDES F 230.00 OPEN BUS 314333 /* ISLAND: 6POWHATN 230.00 OPEN BUS 923862 /* ISLAND: AB2-026 E 230.00 END

11 Light Load Analysis

Not Applicable.

12 Short Circuit Analysis

The following Breakers are overdutied

None.

13 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase.

14 Affected Systems

14.1 TVA

TVA Impacts to be determined during later study phases (as applicable).

14.2 Duke Energy Progress

Duke Energy Progress Impacts to be determined during later study phases (as applicable).

14.3 MISO

MISO Impacts to be determined during later study phases (as applicable).

14.4 LG&E

LG&E Impacts to be determined during later study phases (as applicable).

15 Attachment 1: One Line Diagram and Project Site Location



